

Song-Ze Yu

E-mail: vaclis@berkeley.edu | Personal Page: <https://vaclis.net/> | GitHub: github.com/vaclisinc

Research Interests

My research asks how music AI can augment rather than replace human creativity. I focus on three missing pieces: **controllability**, so musicians can iteratively shape outputs without losing prior intent; **music understanding**, so audio-language models can reliably perceive musical structure and performance; and **humanity**, so models preserve differences among composers, performers, and listeners instead of collapsing them into one ground truth. My long-term goal is to build models that can hear, understand, and communicate about music, enabling richer interaction among musicians, computers, and audiences.

Education

University of California, Berkeley

Computer Science

- GPA: 3.9 / 4.0

Aug. 2025 - May 2026

Advised by Prof. Trevor Darrell and Carmine-Emanuele Cella

National Tsing Hua University

B.S. in Computer Science

- Overall GPA: 3.91 / 4.30

Sep. 2023 - Jun. 2026

Advised by Prof. Liu Yi-Wen

National Cheng Kung University

B.S. in Computer Science and Information Engineering

- Overall GPA: 4.16 / 4.30; highest department rank: 5 / 113

Sep. 2022 - Jun. 2023

Publications

(* indicates equal contribution; † indicates corresponding author)

- [1] **Song-Ze Yu**[†], Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella[†], “*InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement*,” DAFx 2026 Demo, accepted.
- [2] Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corbyn Thomas-Smith, David M. Chan, Karina Nguyen, “*PitchBench: Measuring Pitch Hearing in Audio-Language Models*,” NeurIPS 2026 D&B, under review.
- [3] Milan Liessens Dujardin, **Song-Ze Yu**, “*MuNo-LM: Unifying Score and Performance for Audio-Language Modeling*,” TISMIR, in preparation.
- [4] **Song-Ze Yu**, Joseph Suh, Serina Chang, David M. Chan, “*Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation*,” EMNLP 2026 Demo, under review.

Research Experience

Music and AI Lab, National Taiwan University

Research Assistant

Jun. 2026 - Present

Advised by Prof. Yi-Hsuan Yang

- Improving **controllability in music generation** through explicit music understanding.
- Post-training audio-language models to extend **PitchBench** from evaluating musical perception to teaching models to hear and understand music.

Berkeley AI Research (BAIR) Lab, UC Berkeley

Undergraduate Researcher

Feb. 2026 - Present

Advised by David M. Chan and Prof. Trevor Darrell

- Co-developed **PitchBench**, a 28-experiment hierarchical benchmark showing that frontier audio-language models remain unreliable at fundamental pitch perception across musical and acoustic conditions.
- Built **Anamnesis**, an open-source survey-simulation platform using 30K+ narrative backstories; reproduced Pew Research and New Yorker studies and better matched real-world opinion distributions than persona-prompting baselines.

Center for New Music and Audio Technologies Lab, UC Berkeley

Undergraduate Research Lead

Sep. 2025 - Jun. 2026

Advised by Prof. Carmine-Emanuele Cella

- Formulated **sequential FX refinement** as a stateful multi-turn problem and built **InstructFX2FX**, pairing LLM planning with CLAP-guided optimization to update effect chains without losing prior instructions; outperformed LLM reprompting on 9/10 SocialFX pairs.
- Proposed **MuNo-LM (Music Notation for Language Models)**, a compact representation of performance-level MIDI and score-level intent that enables zero-shot, low-level, note-level, time-grounded caption and QA generation.

AHG Music Lab, National Tsing Hua University

Undergraduate Researcher

Jul. 2024 - May 2025

Advised by Prof. Yi-Wen Liu

- Trained an audio-to-EQ model on a ReaScript-generated piano dataset for tone replication (MSE: 0.0216).

Selected Industry Experience

BerkeleyTime <i>Machine Learning Pod Lead & Full-Stack Engineer</i>	Sep. 2025 - May 2026 Berkeley, CA
• Led development and launch of BerkeleyTime’s new Explore page for 122K+ monthly users; managed a 10-member ML Pod across technical strategy, engineering execution, task allocation, and code reviews. • Built BGE-based semantic search as a FastAPI microservice, then led its migration from FAISS to Redis and MongoDB Vector Search for scalable production deployment. • Reduced GraphQL first-load latency from 25 seconds to 1 second through pagination and Redis caching; shipped 20+ pull requests to a platform used by 30K+ Berkeley students.	
Positive Grid <i>Machine Learning Intern</i>	Jun. 2025 - Aug. 2025 Taipei, Taiwan
• Served as Scrum Master for Amp-AI-SaaS, defining the agentic workflow and system architecture for Positive Grid’s audio products. • Built two JUCE-based VST plugins and integrated an audio-to-preset model through a PyTorch inference pipeline. • Explored source separation and zero-shot virtual-amplifier modeling for music-industry applications.	

Awards & Leadership

Claude × a16z × Berkeley M.E.T. Makeathon — Best Design	Oct. 2025
• Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.	
Meichu Hackathon — 2nd Place, Google Group	Oct. 2024
• Built Travelity, an AI travel planner using interests, personality, and real-time map and search data.	
Taipei Hackathon — 3rd Place	Sep. 2024
• Built ParkFlow, selected for integration into Taipei’s official city app (3M+ downloads); received NT\$100,000.	
SITCON Hackathon — Group Award Winner	Jul. 2024
• Built ccDiary, an AI-assisted mental-health journal with emotion analysis and anonymous social features.	
National Cheng Kung University — Campus Speaker	Aug. 2023
• Delivered an invited talk for incoming students and turned audience-submitted stories into improvised songs performed live.	

Selected Music Experience

Pianist, Composer & Producer	YouTube (7K subscribers)
• Performed 24 solo piano recitals across Greater China and appeared on 100+ Chinese-language TV variety shows and New Year galas between 2009 and 2016. • Shared the stage with Jay Chou three times and appeared as a special guest at Yan Jue’s Taipei Arena concert, performing eight-hands piano with Bai-An and Cosmos People. • Composed 20+ original songs; performs on piano/keys, lead vocal, cajon, and electric guitar, and arranges, mixes, and masters original work.	